

# THE STARSHIP GALILEO

## DEEP SPACE TRAVEL AND COLONIZATION

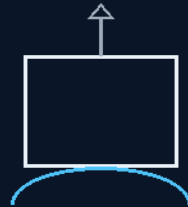
### INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

# PHASE 222

## THE QUINN FLOUR CARTRIDGE

no powder-air interface, ever  
the last powder falls



wet-first doctrine — the capture domes

SEALED CARTRIDGE · CARDBOARD COMPOSITION · GALLEY RULE COMPLETE

**THE DUST NEVER GETS AN INTERFACE**

Galley & particulate — v1.0 in force

**GP-IM-222**

Revision v1.0 — 24 August 2026

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**VETTED BATCH 30 — PASS**

**PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD**

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

## THE GEMINI PROTOCOLS — CHAPTER 222

**PHASE 222: THE QUINN FLOUR CARTRIDGE — PUNCTURE-ON-SCREW SEALED POWDER DISPENSING (WET-FIRST DOCTRINE, CAPTURE DOMES; THE LAST POWDER FALLS; COMPLETES THE GALLEY RULE) — STATUS: CURRENT — PASS; FLOW RATE AND NOZZLE CLOGGING ARE ORDINARY ENGINEERING TESTS.**

The last powder aboard gets its cartridge. Flour was the one dry powder the galley rule could not delete — and flour dust is the stuff that blew up mills. The answer: a cardboard-composition cartridge, oversized to a one-inch screw fitting, its seal punctured BY the screwing of the nozzle — never cut open, never exposed. The gun is shelf tech: 40mm pneumatic cartridge guns already exist for commercial bonding glues. And the doctrine that makes it safe: wet-first, always — flour dispenses only into an already-rotating, lipped bowl already holding the fluid component. Every kitchen machine works under its own open-fronted capture dome. No powder-air interface, ever.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-222, v1.0 — 24 August 2026.

### §0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the cartridge law as seated (contents line, the doctrine — the cartridge, the puncture seal, the shelf gun, wet-first, the dome) — §2 the physics, graded — §3 the system in the cartridge — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

### §1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 222: **The Flour Cartridge — Puncture-on-Screw Sealed Powder Dispensing (Wet-First Doctrine, Capture Domes; the Last Powder Falls; Completes the Galley Rule)**'

THE CORE OBJECTIVE (verbatim): 'Core Objective: powder dispensing with no powder-air interface, ever — sealed cartridge, puncture-on-screw nozzle, wet-first dispensing, capture dome over every kitchen

machine.'

Galley & Particulate Baseline: the galley rule (221, Amendment 3) excluded dry sugar, salt and pepper — flour was the bulk powder still aboard, and flour dust is the founding member of the combustible-dust class: the Washburn A Mill, Minneapolis, 1878, eighteen dead, half the city's milling capacity gone, from dust and a spark. The cartridge deletes the dust interface entirely.

Live Instruction Confirmed: Yes — 1 August 2026, late sitting; declared 'new invention' by the Author on delivery.

### **Live instruction, 1 August 2026.**

"the flour cartridge shall be a cartridge of cardboard composition, with a standard design but oversized to one inch screw fitting for a nozzle with a spring cap"

"the screw end unlike commercial cartridge there tip cut and nozzle fiddle will be punctured by the screwing of the nozzle"

"40mm pneumatic cartridges and guns already exist for commercial bonding glues"

"the flour shall only be dispensed into an already rotating lipped mixing bowl containing any fluid component of the mix, egg milk and eggs or oil or water, as the mixer as with all kitchen machine operate under separate open fronted domes with fluid dynamic air control, there should be zero dust particulates escaping into the kitchen"

### **THE DOCTRINE — AS GIVEN, STRUCTURE READ (KIMI) (VERBATIM)**

- **The cartridge.** Cardboard-composition body (the 221 moulded-fibre line can make them aboard — the feedstock loop closes in the kitchen), standard sausage-pack geometry oversized to a one-inch screw fitting, nozzle with a spring cap: sealed until mated, self-sealing when unmated.
- **Puncture-on-screw.** Unlike commercial cartridges — tip cut open, nozzle fitted, contents exposed — this seal is punctured BY the screwing of the nozzle. No cut tip, no open face, no moment where powder meets air outside the machine. The cream-charger and SodaStream bulb mechanism, doing a baker's job: the fitting is the key.
- **The gun is shelf tech.** 40mm-class pneumatic cartridges and guns already exist for commercial bonding glues — the dispenser is not an invention, the powder cartridge and its doctrine are.
- **Wet-first, always.** Flour dispenses ONLY into an already-rotating, lipped mixing bowl already holding the fluid component — egg, milk, oil, water. The powder lands in moving liquid and is wetted on contact; it never exists as airborne dust because it never has an air interface. The lip catches splash; the rotation carries captured flour under the surface.
- **The dome.** Every kitchen machine operates under its own open-fronted dome with fluid-dynamic air

control — capture-hood class airflow, open face for access, inward flow for containment. Cartridge + wet-first + dome = engineered zero: no particulate path from bag to bowl to kitchen air.

#### **PHYSICS NOTE — THE AUDIT (KIMI: AFFIRMED — THE DUST NEVER GETS AN INTERFACE; TWO BENCH NOTES) (VERBATIM)**

Every layer of the stack attacks the same point — the powder-air interface — and there are no layers left over. Pneumatic sausage guns in the 40-50mm class are mature commercial kit (adhesives, sealants, bonding glues), so the drive train is off the shelf; the puncture-on-screw seal is the proven gas-bulb mechanism, which also makes the cartridge tamper-evident and impossible to 'leave open'; the spring cap closes the nozzle the moment it unmates. Wet-first dispensing is affirmed on the physics: dust is a powder-air phenomenon, and particles that land in moving liquid wet out in milliseconds — there is no airborne phase to capture. The dome is the belt-and-braces layer (ductless capture-hood class), and with the first two layers doing the real work its job is the exception, not the rule — which is exactly how containment should be layered. The flour-dust hazard it deletes is not theoretical: mill explosions wrote the silo law before silos did, and the Washburn A Mill of 1878 remains the record's standing lesson — 218's amendment and the galley rule now have their final member. TWO BENCH NOTES. ONE — the cardboard body wants a food-grade barrier liner, and 221's sheet already seats it: PLA film, grown aboard. TWO — 'zero escape' is an engineered claim, so verify it like one: particle-counter commissioning on every dome, and the count goes in the galley's log. The empty cartridge rides the 221 loop home — compost and re-pulp, not waste. ('cartidrg', 'carboard', 'thare tip cut and nozzle fittle' — reads 'their tip cut and nozzle fitted' — and 'egg milk and eggs' kept verbatim; the readings are standard.)

#### **ORIGIN OF THOUGHT (VERBATIM)**

The galley rule deleted the powders from the manifest — and the Author immediately asked the NP question about the one that couldn't be deleted: bread needs flour, so how does flour exist aboard without ever being dust? The answer is a glue gun with a baker's conscience: the same pneumatic cartridge the adhesives industry has used for decades, but sealed until the nozzle's own screw-thread punches it open, and a law that the only place it may open is into moving liquid, under a dome that catches whatever the law missed. Declared a new invention on delivery, and the audit agrees: the gun is shelf tech, but the sealed powder cartridge and the wet-first doctrine around it are the file's own — and they complete the galley rule. The last powder has fallen.

**Why it matters, in plain English:** Flour was the one dry powder still allowed aboard, and flour dust is the stuff that blew up flour mills on Earth. This cartridge means flour never touches open air: it ships as a sealed cardboard tube with a one-inch screw fitting; screwing the nozzle on is what punctures the seal — there is no cut tip and no open end, ever. A spring cap closes it the moment you unscrew. It clips into a

standard pneumatic gun (the kind glue factories already use), and it is only ever squeezed into a mixing bowl that is already spinning and already holds the wet ingredients — the flour lands in liquid and disappears into it. The mixer, like every kitchen machine, works under its own open-fronted air-curtain dome as the final catch. From store to dough, the flour never gets a chance to become dust.

Build the Flour Cartridge — the cheap impactor testing bed: bags of flour dropped or fired to study cratering and dust behaviour before risking a real penetrator. Cheap, readable, expendable. —

INSTRUCTION RECONSTRUCTION (NOT VERBATIM — the original order was not preserved; reconstructed from the seated body, 10 August 2026, per sitting 146).

## §2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE STACK ATTACKS ONE POINT, AFFIRMED: every layer — sealed cartridge, puncture-on-screw, pneumatic delivery, wet-first bowl, capture dome — removes the powder-air interface, and there are no layers left over. Dust explosion physics needs the interface; the design declines to provide one.

PUNCTURE-ON-SCREW IS THE INVERSION THAT MATTERS: commercial cartridges are tip-cut and nozzle-fitted — contents exposed at every change. This seal is punctured by the act of fitting the nozzle: the powder meets the world only inside the delivery path. Declared 'new invention' by the Author on delivery; the audit found the inversion clean.

THE GUN IS SHELF TECH, AFFIRMED: 40mm-class pneumatic cartridges and guns exist now for bonding glues — sausage-pack delivery, metered, mature. The dispenser is procurement, not invention; the invention is the seal and the doctrine.

WET-FIRST IS THE SAFETY LAW, AFFIRMED: powder enters an already-contained wet mixing environment — no exposed airborne cloud can form, because the target is liquid and rotating. The order is the engineering.

THE DOME CLOSES THE STACK: each kitchen machine under its own open-fronted dome with fluid-dynamic air control — capture-hood class — so even the bowl's own splash-back stays inside the designed air.

## §3 — THE SYSTEM IN THE CARTRIDGE

- **The cartridge:** cardboard-composition body — the 221 moulded-fibre line makes them aboard; the feedstock loop closes in the galley.
- **Puncture-on-screw:** the seal breaks only as the nozzle seats — never cut, never exposed.
- **The shelf gun:** 40mm pneumatic cartridge guns, commercial bonding-glue class — buy the muscle, invent the seal.

- **Wet-first, always:** dispense only into the rotating, lipped bowl already holding egg, milk, water — the fluid first.
- **The dome:** every machine under its own open-fronted capture dome — the air itself is designed.

#### §4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The galley rule (221, Amendment 3) excluded dry sugar, salt and pepper — flour was the bulk powder that could not be deleted, so it got engineered instead: the exception that got its own phase.
- The 221 loop closes twice here: the cartridge body is moulded fibre from the feedstock line, and the wet-first doctrine is the refinery's own unit-dose logic in the kitchen.
- Declared 'new invention' on delivery and the declaration survived the audit — the puncture inversion is the novelty, the gun is the shelf.

#### §5 — VALIDATION: the grade and its full record live in the master and the restart.

#### §6 — BUILD SEQUENCE

- 1. Tool the cartridge: cardboard-composition body, one-inch oversize screw fitting, the puncturable seal.
- 2. Fit the nozzle: the puncture-on-screw fitting — seal breaks only on seating.
- 3. Procure the gun: 40mm pneumatic cartridge gun, bonding-glue class, metered trigger.
- 4. Write the galley law: wet-first sequencing, lipped rotating bowl, dome-on before powder.
- 5. Bench the ordinary tests: flow rate, nozzle clogging, seal puncture reliability, dome capture.

#### §7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 221 (the feedstock question):** the cartridge body from the moulded-fibre line; the galley-rule exclusion that created this phase.
- **Phase 235 (the vac pack):** the bay-door kitchen's sibling — powder handled by cartridge, produce handled by slide-tray; one air-discipline doctrine.
- **Phase 198 (the cake oven):** the bowl this cartridge feeds — the rotisserie line's wet-first partner.

## **§8 — DO-NOT-BUILD**

- Do NOT cut the tip and fit the nozzle — commercial practice is the exposure this phase deletes; puncture-on-screw only.
- Do NOT dispense into a dry bowl — wet-first is law; the order is the safety case.
- Do NOT run powder outside the dome — the capture dome is part of the machine, not an accessory.
- Do NOT invent a dispenser — the gun is shelf tech; invention effort goes to the seal and the doctrine.
- Do NOT treat clogging as a failure mode surprise — it is the named ordinary test; bench it early.

## **§9 — OWED**

OWED: the ordinary engineering tests the vet named — powder-flow rate and nozzle clogging — plus seal puncture reliability across production variance, dome capture verification at full mix speed, and the cartridge body's moisture behaviour from the 221 fibre line.

## **§10 — STATUS / PROVENANCE / CHANGE CONTROL**

STATUS: CHAPTER GP-IM-222, v1.0 — CURRENT — PASS; FLOW RATE AND NOZZLE CLOGGING ARE ORDINARY ENGINEERING TESTS, seated law, master v2.1 (numerical print order). Built 24 August 2026, third of the 20-chapter 'ok ill do these 20, you do 20 more' shift (the author's order, 24 August 2026, seated in sitting 618). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607 and 618): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 222